

Headlines Completion Results

2024-10-24

Headlines Completion

- **Number of Observations:** 20,000.
- **Cleaning:** For responses missing the provided part of the headline, we append it. If a response starts differently from the provided headline, we leave it as is. Both the true and LLM headlines are cleaned by removing **non-alphanumeric** characters and converting everything to lowercase.
- **Embedding Model:** Cleaned headlines are embedded using OpenAI’s `text-embedding-3-small` model, with an embedding size of 1536.

Examples of Successful Completion

Here are 10 examples where GPT-4o is considered to successfully complete the bill summary. Specifically, the cleaned responses are identical to the cleaned true summaries.

Table 1: Examples of Successful Completion of Headlines (Non-Embedded)

Model	Prompt	HeadlineID	HeadlineTrim	HeadlineClean	HeadlineLLMClean
GPT-4o	w/o date restric- tion	14220	China Said To Offer US Concessi	china said to offer us concessions to eliminate trade imbalance	china said to offer us concessions to eliminate trade imbalance
GPT-4o	w/o date restric- tion	41728	ChinaNet Online Names Centurion ZD CPA & Co. A	chinanet online names centurion zd cpa co as independent registered public accounting firm	chinanet online names centurion zd cpa co as independent registered public accounting firm
GPT-4o	w/o date restric- tion	4155	Piper Jaffray Maintains Overweight on Acti	piper jaffray maintains overweight on activision blizzard raises price target to 62	piper jaffray maintains overweight on activision blizzard raises price target to 62
GPT-4o	w/o date restric- tion	32403	Nomura Maintains Buy on FMC,	nomura maintains buy on fmc raises price target to 100	nomura maintains buy on fmc raises price target to 100

Table 1: Examples of Successful Completion of Headlines (Non-Embedded) (*continued*)

Model	Prompt	HeadlineID	HeadlineTrim	HeadlineClean	HeadlineLLMClean
GPT-4o	w/o date restriction	34422	Sinclair Completes Acquisition of R	sinclair completes acquisition of regional sports networks from disney	sinclair completes acquisition of regional sports networks from disney
GPT-4o	w/ date restriction	41546	Charter Late Tuesday Priced \$500M	charter late tuesday priced 500m of senior unsecured notes due 2030	charter late tuesday priced 500m of senior unsecured notes due 2030
GPT-4o	w/ date restriction	23350	H. Ross Perot, Computer Industry Pioneer An	h ross perot computer industry pioneer and former presidential candidate dies at 89	h ross perot computer industry pioneer and former presidential candidate dies at 89
GPT-4o	w/ date restriction	47844	Euronet Worldwide shares are trading lower after DA	euronet worldwide shares are trading lower after da davidson downgraded the stock from buy to neutral	euronet worldwide shares are trading lower after da davidson downgraded the stock from buy to neutral
GPT-4o	w/ date restriction	12629	Tesla To Open A Gi	tesla to open a gigafactory in berlin	tesla to open a gigafactory in berlin
GPT-4o	w/ date restriction	50841	Report: Comcast Might S	report comcast might sell hulu stake to disney	report comcast might sell hulu stake to disney

Tables and Figures

Table 2: Share of Identical True and GPT-4o Headlines in Non-Embedded Form

Model	Average
GPT-4o w/o date restriction	0.006
GPT-4o w/ date restriction	0.007

Table 3: Cosine Similarity and Euclidean Distance Between True and GPT-4o Embedded Headlines

Model	Average	Baseline
<i>Cosine Similarity</i>		
GPT-4o w/o date restriction	0.880	0.309
GPT-4o w/ date restriction	0.880	0.309
<i>Euclidean Distance</i>		
GPT-4o w/o date restriction	0.455	1.172
GPT-4o w/ date restriction	0.455	1.172

Figure 1: Cosine Similarity Between True and LLM Headlines Compared to a Random Baseline, GPT-4o, by Prompt

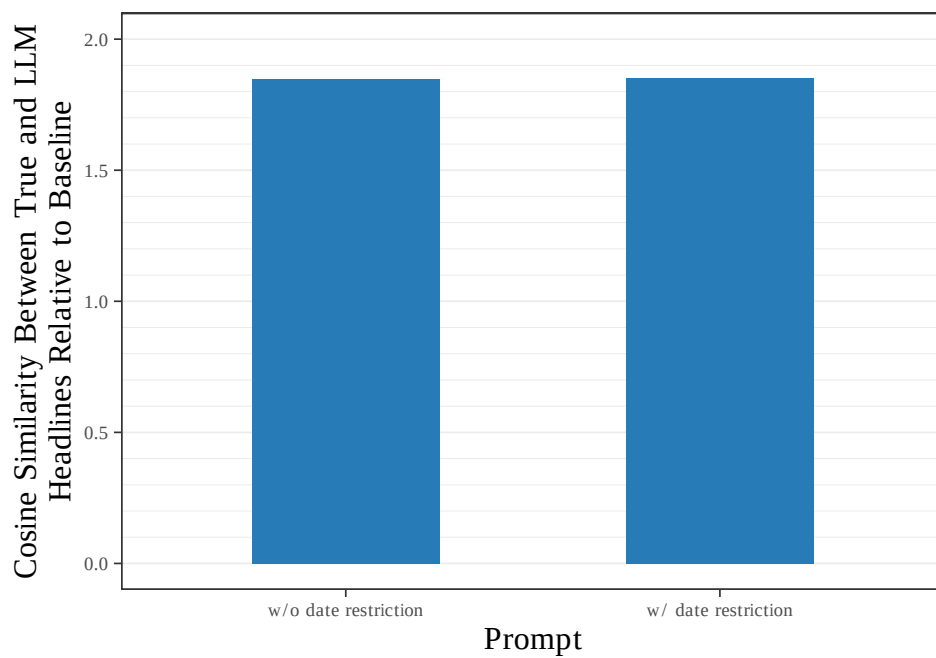


Figure 2: Euclidean Distance Between True and LLM Headlines Compared to a Random Baseline, GPT-4o, by Prompt

